Orbit Social Media Application — Handbook (PDF-ready)

Product: Orbit — a photo-first social web application

Primary surfaces: Feed, Reels, Messages, Profile (plus authentication)

Tech stack (current codebase): Next.js (App Router), Tailwind CSS, Supabase (Auth + Postgres when configured), Vercel hosting

Document purpose: Explain what Orbit is, why it exists, how it is used, how it was created through staged delivery, what the key features are, how it reaches audiences, what issues were encountered, and what to improve next time.

How to export this document to PDF (recommended)

Option A — Microsoft Word / Google Docs (fastest, best layout control)

1. Open this file in an editor that renders Markdown (VS Code preview, GitHub, etc.).
2. Copy the rendered content into **Word** or **Google Docs**.
3. Insert/replace screenshots from `docs/assets/` as needed (drag images into the doc).
4. Add a **Cover page** + **Table of contents** (Word: References → Table of Contents).
5. Export: **File** → **Save As / Download** → **PDF**.

Option B — Pandoc (automated, optional)

If you install Pandoc locally, you can convert Markdown to PDF with a PDF engine (requires a LaTeX distribution or an engine like `wkhtmltopdf`). This repo does not require Pandoc for development.

Cover page content (paste at top of your PDF)

Title: Orbit Social Media Application — Engineering & Product Handbook

Subtitle: Feed, Reels, Messages, Profile + Authentication (Supabase + Vercel)

Author: Baljinder Singh Rai / BaljinderSingh (team)

Repository: `balisikh/orbit-social-media-application`

Deployment platform: Vercel

Backend platform: Supabase

Date: April 2026

Live URL (current): `https://orbit-social-media-application.vercel.app/`

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1) Executive summary

Orbit is designed as a modern social product with four core modules:

- **Feed** for browsing posts.
- **Reels** for short vertical video.

- **Messages** for direct conversations.
- **Profile** for identity and a portfolio grid of posts.

The application is engineered to run locally for development, and to deploy to production using **Vercel**. In the current deployment, Supabase is not configured and Orbit runs in **Local mode** (browser-only data) so users can try the full UI without a backend.

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2) What Orbit is (definition)

Orbit is a **dynamic website** (server-rendered routes + client interactions) that provides social-media workflows:

- **Authentication:** sign up and sign in (Supabase email/password when configured; **Local mode** browser-only sessions when not configured).
- **Profile identity:** display name, username/handle, bio, avatar.
- **Content publishing:** posts (including slideshow-style multi-media posts in local mode paths) and reels.
- **Social graph primitives:** follow-related flows exist in the codebase for Supabase-backed usage (depending on migrations and configuration).

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3) Why Orbit exists (purpose)

Product purpose

- Provide a **cohesive** experience across Feed/Reels/Messages/Profile.
- Prioritize **photo-first** consumption patterns (feed + profile grids).
- Support a realistic path to production hosting (Vercel) with a real backend (Supabase).

Engineering purpose

- Keep UI responsive across **mobile, tablet, and desktop** using Tailwind breakpoints (`sm` , `md` , `lg` , ...).
- Keep authentication and session refresh predictable in production.
- Separate **local preview** development behavior from **production** behavior.
- Support a **Local mode** deployment on Vercel (no Supabase) for demo/portfolio use, where data is saved per-browser.

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4) How Orbit is used (user journeys)

Journey A — New user onboarding

1. User visits `/auth/signup` .
2. User creates an account (Supabase or Local mode).
3. User completes profile basics on `/me/edit` (display name, @handle, bio, avatar).
4. User creates a first post on `/me/post/new` .
5. User sees content on `/feed` and `/me` .

Journey B — Returning user

1. User visits `/auth/login` .
2. User lands on `/feed` .

3. User checks `/reels` and `/messages`.
4. User visits public profile `/u/[handle]` (when handle exists).

Journey C — Password recovery

1. User visits `/auth/forgot-password`.
2. Supabase sends reset email (only when Supabase is configured; Local mode does not support password reset emails).
3. User completes `/auth/update-password` after the callback flow.

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5) Stages and process (create → design → develop → test → deploy)

Stage 1 — Concept and scope

Outputs: feature list, MVP boundaries, success metrics.

Orbit MVP modules captured in this repo: Feed, Reels, Messages, Profile, Auth.

Stage 2 — UX and information architecture

Outputs: route map and primary navigation.

Primary routes (App Router):

- `/feed`
- `/reels`, `/reels/new`
- `/messages`
- `/me`, `/me/edit`, `/me/post/new`
- `/u/[handle]`
- `/auth/login`, `/auth/signup`, `/auth/forgot-password`, `/auth/update-password`, `/auth/callback`

Stage 3 — System design

Outputs: data model direction, auth strategy, deployment strategy.

Key decisions reflected in the codebase:

- **Supabase** for cloud auth and persistence when `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY` are configured.
- **Vercel** for hosting and continuous deployment from GitHub.
- **Local mode fallback (no Supabase):** allows sign in + full UI demos on Vercel with browser-only storage.

Stage 4 — Development

Outputs: implemented UI + API routes + database migrations (SQL under `supabase/migrations/`).

Engineering notes from this project lifecycle:

- Next.js 16 introduced a rename from root `middleware` convention to **`proxy`** convention for edge/session refresh patterns.
- Responsive rules were tuned for:
- **Portrait single-image posts** vs **slideshow posts** in the feed.
- **Profile grids** across breakpoints.

Stage 5 — Testing

Minimum production test checklist:

- Auth: signup/login/logout/forgot password (as applicable)
- Profile: `/me` and `/me/edit` persistence
- Feed: post creation and rendering
- Reels: list + upload flow (as configured)
- Messages: inbox behavior (mode-dependent)
- Public profile: `/u/[handle]`

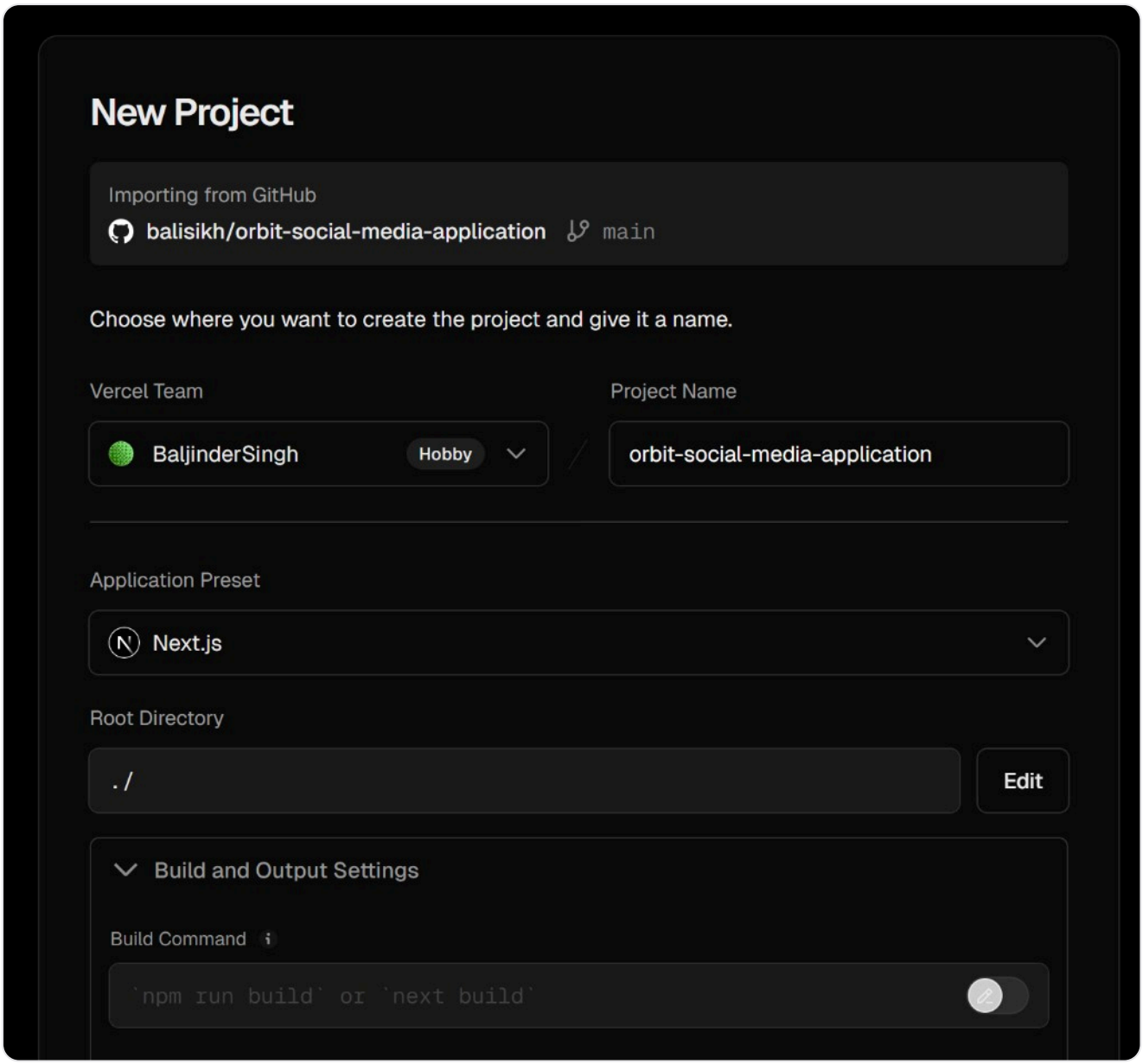
Responsive test widths (examples):

- Mobile: 360×800, 390×844, 412×915
- Tablet: 768×1024, 834×1194, 1024×1366
- Desktop: 1366×768, 1440×900, 1920×1080

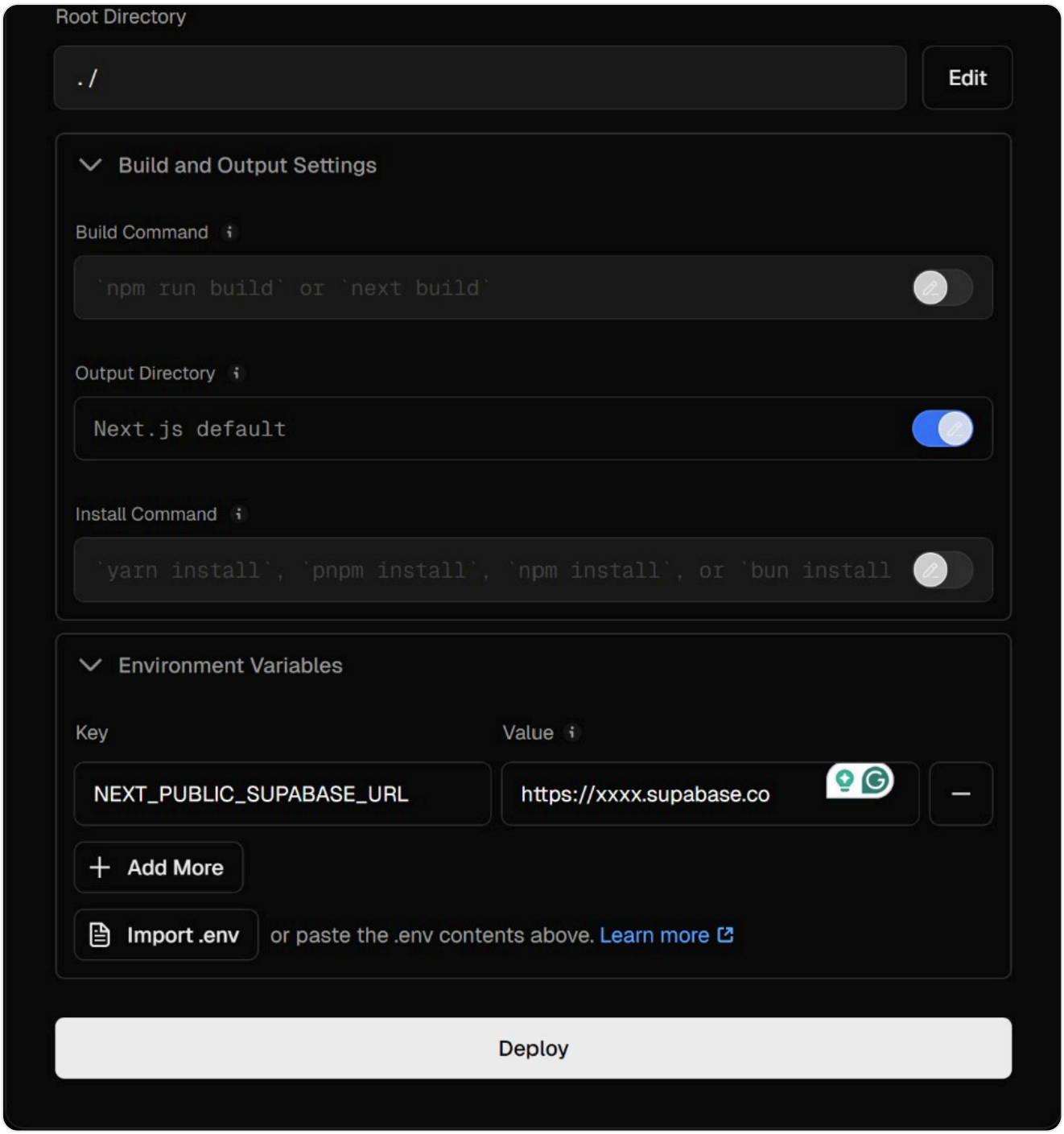
Stage 6 — Deployment (Vercel)

Outputs: live URL, environment configuration, monitoring.

Screenshots included in this handbook (from your deployment flow):



Vercel new project import from GitHub



Vercel new project environment variables and deploy

Important deployment note:\n+\n+- For **Supabase mode**, ensure `NEXT_PUBLIC_SUPABASE_URL` is your real Supabase URL (not a placeholder), and add `NEXT_PUBLIC_SUPABASE_ANON_KEY`.\n+- For **Local mode** (current live deployment), Supabase is intentionally not configured, and all data is stored in each browser.\n+\n+**Live deployment (current):** `https://orbit-social-media-application.vercel.app/`

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6) Key features (what / why / how)

6.1 Feed (`/feed`)

- **What it is:** A scrolling stream of posts.
- **Why it exists:** Primary daily-use surface for social content consumption.

- **How it is used:** Users scroll posts, interact with actions provided by the UI (likes/comments/share vary by mode), and navigate to profile-related flows.

Content-dependent UI behavior (important for documentation accuracy):

- **Portrait photos + short captions:** framed for consistent card layout.
- **Slideshow posts:** framed for swipe/arrow navigation with consistent height on larger breakpoints.

6.1.1 Feed — what / why / how / ways (expanded)

- **What it is**: The main scrolling stream where posts appear as cards. Each card can be a single portrait photo, a slideshow (multiple media items), or a video.
- **Why we have it**: This is the core "daily open" surface—users come here to catch up quickly.
- **How we use it**:
- Open **Feed** → scroll posts
- Use actions: **Like**, **Comment**, **Share**
- Tap **New post** to publish content
- **Ways of doing it (modes)**:
- **Local mode (current Vercel demo)**: posts are stored in the browser under `orbit:posts:<email>`. Each device/browser has its own feed.
- **Supabase mode (optional production)**: posts and follow graph are stored in a database so feed content is shared across users/devices.

6.2 Reels (`/reels` , `/reels/new`)

- **What it is:** Vertical short video browsing and publishing entry points.
- **Why it exists:** Short video is a distinct consumption pattern from static feed posts.
- **How it is used:** Users browse reels tabs and upload reels.

6.2.1 Reels — what / why / how / ways (expanded)

- **What it is**: Full-height vertical video content in a swipeable player.
- **Why we have it**: Reels support discovery and fast entertainment, separate from the photo feed.
- **How we use it**:
- Open **Reels** → watch and swipe
- Create reels via **New reel**
- **Ways of doing it (modes)**:
- **Local mode**: reel metadata is stored in localStorage (`orbit:reels:<email>`), while video bytes are stored in IndexedDB (so it doesn't hit localStorage quota quickly).
- **Supabase mode**: reel metadata goes to DB and videos go to storage (not enabled in Local-only deployments).

6.3 Messages (`/messages`)

- **What it is:** Direct messaging UI.
- **Why it exists:** Completes the social loop beyond public posting.
- **How it is used:**
- With Supabase configured: server-backed messaging paths are used.
- Without Supabase (development preview): local browser preview behavior may be used for demos.

6.3.1 Messages — what / why / how / ways (expanded)

- **What it is**: A split inbox + thread layout with a composer, attachments (gallery/stickers), and reactions.
- **Why we have it**: Direct conversation is a core social feature—users connect privately after discovering content.
- **How we use it**:
- Open **Messages** → select a conversation → send messages
- Attach photo/video from gallery and add reactions
- **Ways of doing it (modes)**:
- **Local mode**: threads are stored in the browser under `orbit:messages:<email>`.
- **Supabase mode**: messages are stored in the database with RLS policies.

6.4 Profile (`/me`, `/me/edit`, `/u/[handle]`)

- **What it is:**
- `/me` is the signed-in owner profile hub.
- `/u/[handle]` is the public profile page for a handle.
- **Why it exists:** Identity + portfolio + shareable profile link.
- **How it is used:** Users set display name, username/handle, bio, avatar; visitors can follow/message depending on configuration.

6.4.1 Profile / follow graph — what / why / how / ways (expanded)

- **Profile basics (name, @handle, bio, avatar)**
- **What it is**: Your identity panel.
- **Why we have it**: People need a consistent identity and shareable profile link.
- **How we use it**: `/me/edit` → set photo/name/@handle/bio → Save.
- **Ways of doing it**:
- **Local mode**: saved per browser (avatar stored at `orbit:avatar:<email>`, and profile basics in a browser-only session backup).
- **Supabase mode**: stored in DB + auth metadata and shows everywhere on login.
- **Following**
- **What it is**: Accounts you follow.
- **Why we have it**: Following determines what should show in your feed and builds the social graph.
- **How we use it**: open **Following** tab → view/unfollow.
- **Ways of doing it**:
- Local mode: stored in `orbit:follows:v2` in localStorage.
- Supabase mode: stored in DB, enforced by RLS.
- **Followers**
- **What it is**: People who follow you.
- **Why we have it**: Audience + safety controls.
- **How we use it**: open **Followers** tab → search → remove/block.
- **Ways of doing it**:
- Organic: someone visits `/u/[handle]` and taps Follow.
- Demo: use **Demo tools** to generate requests and followers for testing.
- **Requests (follow requests)**

- **What it is**: A queue of people requesting to follow you.
- **Why we have it**: Supports approval workflows and prevents unwanted followers.
- **How we use it**: open **Requests** → **Accept** or **Decline**.
- **Ways of doing it**:
- Demo tools can bulk-generate requests to test the UI.
- **Block**
- **What it is**: Prevents an account from following/interacting.
- **Why we have it**: Safety and control.
- **How we use it**: Followers list → **Block**; Blocked list → **Unblock**.
- **Demo tools**
- **What it is**: Tools to generate demo requests/followers to make the profile look realistic for testing.
- **Why we have it**: Lets you test the end-to-end UI without waiting for real users.
- **How we use it**: Demo tools → enter a count → Add requests → manage them in Followers/Requests.

6.5 Authentication (`/auth/login`, `/auth/signup`, password reset pages)

- **What it is:** Email/password authentication via Supabase when configured; otherwise Local mode creates a browser-only session so users can use Orbit without a backend.
- **Why it exists:** Secure accounts and server-backed personalization.
- **How it is used:** Users sign up/sign in; sessions are maintained for app routes. In Local mode, sessions and data persist in browser storage for that domain.

6.5.1 Authentication — what / why / how / ways (expanded)

- **What it is**: The entry point to identify a user, so Orbit can load the right profile/posts/messages.
- **Why we have it**: Without identity, Orbit can't know which data belongs to which user (even in Local mode).
- **How we use it**:
- `/auth/signup` creates a session
- `/auth/login` restores a session
- `/auth/forgot-password` and `/auth/update-password` are available only in Supabase mode (email workflow)
- **Ways of doing it**:
- **Local mode**: creates a browser-only session cookie and stores user data locally per domain.
- **Supabase mode**: creates a Supabase session (JWT/cookies) and persists data in the database.

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7) Reaching target audiences (distribution and growth)

Who the product is for (example segments)

- **Creators** who want a simple portfolio + posting flow.
- **Friends/community groups** who want feed + DMs.
- **Early adopters** validating UX before scaling features.

How Orbit can reach audiences

- **Shareable public profiles:** `/u/[handle]` links can be posted externally.
- **Invite loops:** "create account → follow → message" flows.

- **Content marketing:** demo reels/posts as examples in feed.

Benefits to users

- **Fast onboarding:** clear module separation (Feed/Reels/Messages/Profile).
- **Cross-device UX:** responsive layouts tuned for common breakpoints.
- **Production viability:** deployable architecture reduces “prototype-only” risk.

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8) Issues encountered (real examples from this project) + causes + mitigations

Issue 1 — “Password reset is not available…” on production

- **Symptom:** reset page shows unavailable messaging.
- **Cause:** Supabase is not configured (Local mode). Password reset requires Supabase email delivery.
- **Mitigation:** either configure Supabase (URL + anon key + redirect URLs + email) or keep Local mode and remove password reset from expectations.

Issue 2 — “Sign-in is not available on this server.”

- **Symptom:** login cannot proceed in production.
- **Cause:** earlier builds blocked sign in when Supabase was missing.
- **Mitigation:** implement **Local mode sign in** when Supabase is not configured (current approach on the live site).

Issue 3 — Profile fields reverting / conflicting with auth metadata defaults

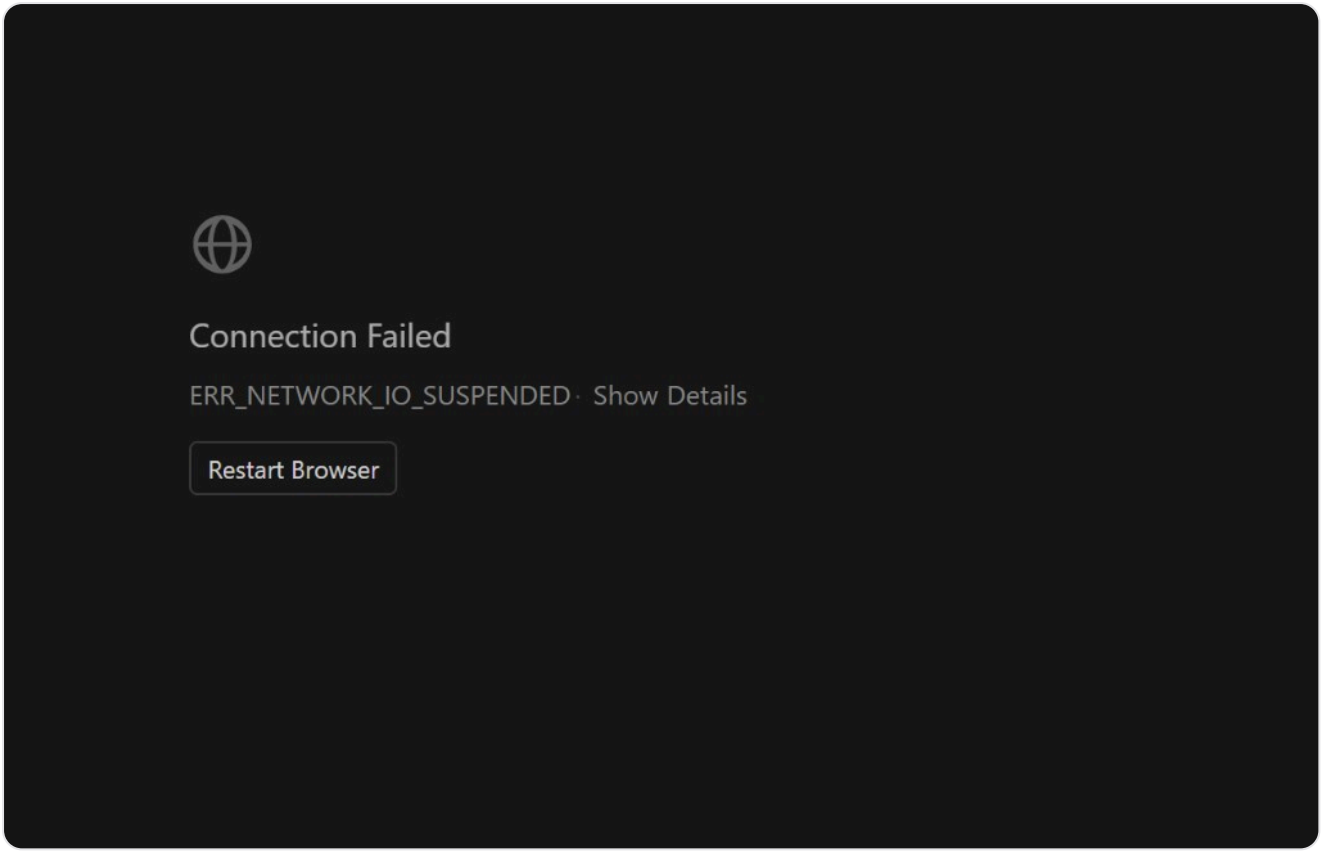
- **Symptom:** profile appears to revert toward email-local defaults after sessions.
- **Cause:** profile sync/merge logic previously favored provider defaults in ways that could overwrite intended profile row values; also confusion between local preview vs cloud mode.
- **Mitigation:** improve merge rules and keep a single “source of truth” strategy per environment; document the difference between preview and production.

Issue 4 — Next.js middleware deprecation warning

- **Symptom:** build warns that middleware file convention is deprecated.
- **Cause:** Next.js 16 renamed the convention to `proxy`.
- **Mitigation:** migrate root `middleware.ts` to `proxy.ts` exporting `proxy`.

Issue 5 — Network connectivity errors during testing

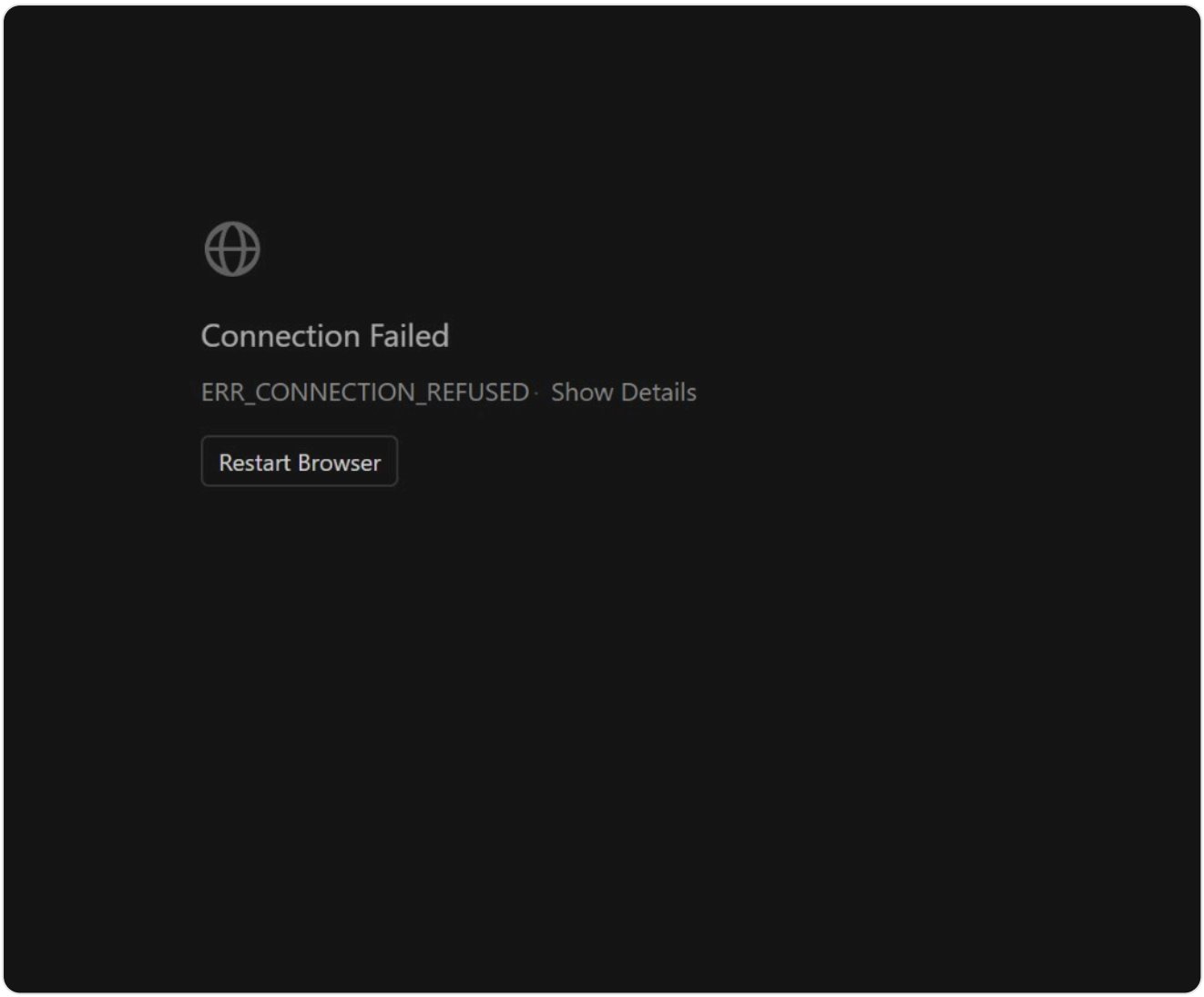
The following screenshot captures a browser-level connectivity failure (`ERR_NETWORK_IO_SUSPENDED`). This is not an Orbit application logic error by itself, but it is a realistic operational issue during local testing or unstable networks.



Connection failed error example

- **Mitigation:** retry on stable network; restart browser; verify VPN/proxy; verify local dev server is running; verify deployed site DNS/SSL on Vercel.

Additionally, a local dev server can show `ERR_CONNECTION_REFUSED` when the Next.js dev server is not running or the wrong URL is used (e.g. opening a Vercel URL as if it were localhost).



Connection refused error example

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9) Screenshots you should add for a “complete” PDF (recommended next captures)

The screenshots included above cover **Vercel project creation + env var entry**. For a complete stakeholder PDF, add these additional screenshots (name them consistently under `docs/assets/`):

1. **Orbit live site** `/feed`, `/reels`, `/messages`, `/me`, `/me/edit`, `/auth/login` (showing Local mode banner)
2. **Data transfer UI** on `/me/edit` (Export/Import local data)
3. **Vercel deployment success** page showing “Ready” and the live domain
4. (Optional, if moving back to Supabase) **Supabase dashboard** → **API keys** and **Auth URL configuration** (blur secrets)

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10) How the final project was produced (end-to-end)

This section summarizes the actual steps used to produce the Orbit Social Media Web Application and get it running locally and on Vercel.

10.1 What was built (final state)

Orbit delivers these modules:

- **Feed**: create and browse posts, including single portrait photos and slideshow posts.
- **Reels**: browse vertical reels and create reels in Local mode (with video stored in IndexedDB).
- **Messages**: inbox + thread UI stored locally in Local mode.
- **Profile**: display name, @handle, bio, avatar, post grid, and local follow tools (followers/following/requests/block + demo tools).
- **Local data transfer**: export/import Local mode data so the Vercel site can match the Cursor/localhost setup.

10.2 How it was created (Cursor workflow)

Typical implementation workflow used:

1. **Run the app locally** with `npm run dev`.
2. **Build UI and flows** across the main routes (`/feed` , `/reels` , `/messages` , `/me` , `/me/edit` , `/me/post/new`).
3. **Improve responsiveness** so the UI works on mobile, tablet, and desktop (Tailwind breakpoints).
4. **Fix production warnings** (e.g., Next.js `middleware` convention renamed to `proxy`).
5. **Verify correctness** repeatedly using:

- `npx tsc --noEmit`
- `npm run lint`
- `npm run build`

1. **Commit and push** updates to GitHub, then deploy through Vercel.

10.3 Deployment steps (Vercel)

These steps describe how the app was deployed from GitHub to Vercel:

1. **Import from GitHub**: select repository `balisikh/orbit-social-media-application` on branch `main` .
2. **Project settings**:

- Framework preset: **Next.js**
- Root directory: `./`
- Build/output: defaults

1. **Choose deployment mode**:

- **Local mode (current live deployment)**: do not configure Supabase keys; Orbit runs with browser-only storage so anyone can try the UI.
- **Supabase mode (optional production)**: set `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY` , apply DB migrations, set Auth redirect URLs, and redeploy.

1. Click **Deploy** and verify the deployment shows **Ready**.

10.4 How to keep Vercel “up to date”

To keep the live site updated with what you build in Cursor:

1. Make changes locally.
2. Run checks: `tsc` , `lint` , `build` .
3. `git commit` and `git push` to `main` .
4. Vercel auto-deploys from GitHub (or you can click **Redeploy** in Deployments).

10.5 How the live site matches the Cursor setup (Local mode)

Because Local mode data is stored per-domain, localhost and Vercel do not automatically share the same profile/posts/follows.

To make them identical:

1. On **localhost**: open `/me/edit` → use **Export data**.
2. On **Vercel**: open `/me/edit` → use **Import** and upload the exported JSON.

This copies local profile basics, avatar, posts, follows/requests/actions, and messages (and some reel media, within size limits).

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10) Conclusion — what to do next time (process improvements)

1. **Decide the deployment mode early:**\n+ - **Local mode** for demo/portfolio (browser-only, no shared data)\n+ - **Supabase mode** for real multi-user use (shared database + auth)\n+2. **Pre-deploy checklist** pinned in the repo README or team wiki (Supabase mode):\n+ - `NEXT_PUBLIC_SUPABASE_URL`\n+ - `NEXT_PUBLIC_SUPABASE_ANON_KEY`\n+ - Supabase redirect URLs include `/auth/callback`\n+ - migrations applied
2. **CI gate:** run `tsc`, `lint`, and `build` on every PR.
3. **Separate "preview vs production" documentation** so testers do not confuse local preview behavior with cloud persistence.
4. **Operational monitoring:** Vercel deployment logs + Supabase auth logs for first-day issues.

Appendix A — Glossary

- **Vercel:** hosting and deployment for the Next.js app.
- **Supabase:** managed Postgres + authentication APIs.
- **RLS:** row-level security policies in Postgres.
- **`NEXT_PUBLIC_*` variables:** bundled into client-side code; must be safe/public values.

Appendix B — File map for screenshots used in this handbook

- `./assets/vercel-new-project-import.png`
- `./assets/vercel-new-project-env-vars.png`
- `./assets/error-connection-failed-network-io-suspended.png`
- `./assets/error-connection-refused.png`

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